GLASGOW COLLEGE UESTC

Homework
Academic Year 2025-2026
Semester II

Power Electronics (UESTC3022)

Attempt all PARTS. Total 100 marks

Power Electronics 2 Formulae List

For an inductor: $v_L(t) = L\dfrac{di_L(t)}{dt}, \quad i_L(t) = \dfrac{1}{L}\int v_L(t)dt, \quad E_L(t) = \dfrac{1}{2}L\left[i_L(t)\right]^2$

For a capacitor: $v_C(t) = \dfrac{1}{C}\int i_C(t)dt, \quad i_C(t) = C\dfrac{dv_C(t)}{dt}, \quad E_C(t) = \dfrac{1}{2}C\left[v_C(t)\right]^2$

Average value: $V_{ave} = \dfrac{1}{T}\int_0^T v(t)dt$

RMS value: $V_{rms} = \sqrt{\dfrac{1}{T}\int_0^T v(t)^2\,dt}$

Duty cycle: $D \equiv \phi = \dfrac{T_{on}}{T_{on}+T_{off}} = \dfrac{T_{on}}{T_{period}}$

Amplitude modulation index of SPWM:

$$m_a = \frac{\hat{V}_{control}}{\hat{V}_{carrier}}$$

Frequency modulation index of SPWM:

$$m_f = \frac{f_{carrier}}{f_{control}}$$

Instantaneous power:

$$p(t) = \frac{v^2(t)}{R} = i^2(t)R = v(t)\times i(t) = force \times velocity = torque \times angular\ velocity$$

Average power:

$$P = \int_0^T v(t)i(t)dt = \frac{V_{rms}^2}{R} = I_{rms}^2 R$$

reverse recovery power loss in a diode:

$$P_{RR} = Q_{RR} \times V_{R(blocking)} \times f_{switching}$$

MOSFET switching loss:

$$P_{switching} = \frac{f_{switching}V_{DS(off)}}{2}\left(T_{on}I_{on} + T_{off}I_{off}\right)$$

Thermal circuit:

$$\Delta T = P_{\text{device}} \times \Sigma\theta_{\text{XY}} \quad \text{or} \quad \Delta T(°\text{C}) = \frac{power \times time}{thermal\ capacitance}$$

Junction temperature of device:

$$T_{\text{junction}} = T_{\text{sink}} + P_{\text{device}}\left(\theta_{\text{juntion-case}} + \theta_{\text{case-sink}}\right)$$

Calculus:

$$\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$$

$$\int \cos\left(a\theta\right)d\theta = \frac{1}{a}\sin\theta \qquad \int \cos\left(a\theta\right)d\theta = -\frac{1}{a}\sin\theta$$

Q1. Answer all questions in this section:

a. For the periodic load current waveform of Figure Q1-a, calculate:

i. The average current. [1]

ii. The RMS current. [2]

iii. The form factor of the waveform. [2]

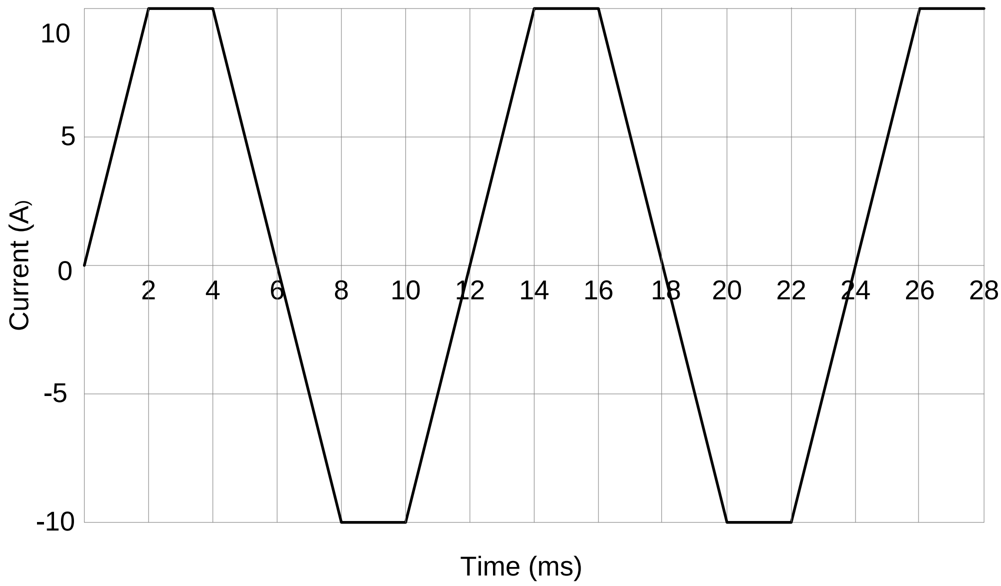

Figure Q1-a

b.

i. Draw and label a circuit to show how two silicon controlled rectifiers (SCRs) can be used to control a resistive load from an AC supply, allowing current to flow in both directions through the load. [2]

ii. Draw and label sketches of the current through the load to show how phase-angle control of SCRs is used to provide an average power (1) less than 50% of its maximum value and (2) above 50% of its maximum value. [3]

c. Please list three advantages and three disadvantages of MOSFETs. [5]

d. A switch-mode power supply contains two power semiconductor devices:

- A MOSFET switch, which dissipates 1.0 W.
- A diode, which dissipates 2.0 W.

Both are in TO220 packages, which have thermal resistances Θ_{jc} =3 °CW^{-1} and Θ_{ca} =60 °CW^{-1}. The temperature of their junctions should not exceed 125°C and the ambient temperature is 25°C.

i. Is it possible to use either device safely without a heatsink? [2]

ii. Despite the results of the previous analysis, the designer decides to mount both devices on a common heatsink. Specify the heatsink. (Hint: which component is most at risk from overheating?) [3]

e. Consider the circuit in Figure Q1-e. Assume the power diode's forward voltage drop = 0.7 V. When an 80 V_{rms} sine voltage is applied to the primary winding of the transformer,

i. What type of circuit is this? [1]

ii. What is the total peak secondary voltage ($V_{p(sec)}$)? [1]

iii. Draw the voltage waveform across R_L [1]

iv. What is the peak current through each diode? [1]

v. Determine the peak inverse voltage (PIV) for each diode. [1]

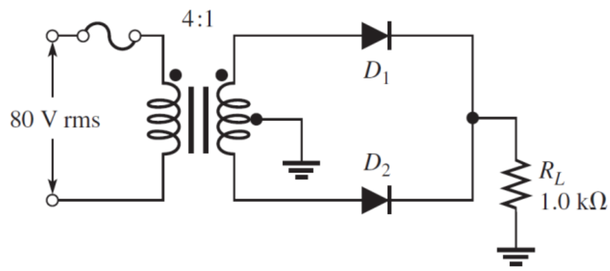

Figure Q1-e

Q2. a. For the full-wave diode rectifier circuit shown in Figure Q2, assuming the input voltage is $V_{AC} = 100\,sin(\,100\pi t)$, the conduction angle of the diode is 30°, the load resistance R_L = 20 Ω, and the capacitor C_S = 10 mF. We assume that all components are perfect and neglect the voltage drop across the diodes.

i. Estimate the peak-to-peak ripple voltage in the output voltage V_R and the peak inverse voltage in the voltage drop V_D for the diode D. [4]

ii. If the output filter capacitor C_s is removed, sketch and label the voltages across the load R_L and the diode D separately, showing the time scale clearly. Calculate the RMS value of the output voltage V_R. [4]

iii. If the ripple voltage across the load resistance R_L must not exceed 1V peak-to-peak. Calculate the value of smoothing capacitance C_S required. You may assume that the capacitor provides current to the load for each complete cycle of the rectified voltage. [4]

iv. If only increase the value of Cs, what will it affect V_R ripple, the power diode conduction angle, the form factor of the current waveform drawn from the transformer, and the VA rating of the transformer, respectively? [4]

v. List the advantages and disadvantages of this full bridge rectifier circuit. [4]

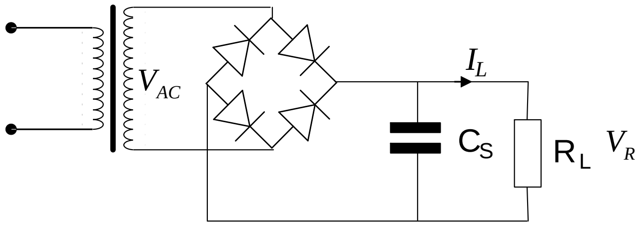

Figure Q2

b. A MOSFET has thermal resistance from the junction to case θ_{jc} as 1.8 °C/W and the thermal resistance from the case to the heat sink θ_{cs} as 0.6 °C/W.

i. If the device is mounted on a heat sink that has a thermal resistance of θ_{sa} 7 °C/W, determine the maximum power that can be absorbed without exceeding a junction temperature of 160 °C when the ambient temperature is 50 °C. [1]

ii. Determine the junction temperature when the absorbed power is 20 W. [2]

iii. Determine θ_{sa} of a heat sink that would limit the junction temperature to 160 °C for 20 W absorbed. [2]

Q3. You are designing a high-speed power converter for a critical portable device that needs to be secured. During testing, you observe high-frequency voltage oscillations across the main switching MOSFET. If not mitigated, these spikes could cause:

Physical Breach: Dielectric breakdown of the MOSFET (Device Failure).

Electromagnetic Compliance and Information Leakage: Radiated Electromagnetic Interference (EMI) that on one side can affect other nearby devices and on the other side can allows a remote attacker to "sniff" the device's operating frequency and power state from a distance. This can allow the sniffer to perform malicious activity of disturbing the function of the critical application.

i. Consider that the parasitic board inductance of Lp = 20 nH and a device stray capacitance of 500 pF, the switching frequency of 200 KHz and the bus voltage of 400V, design an RC Snubber Circuit to mitigate these risks. [8]

ii. Calculate the power loss introduced by this snubber circuit. Evaluate the significance of this loss in the context of a portable device. Select and apply appropriate a compromise between oscillation suppression and minimising power loss. [8]

iii. Do a literature survey and cite any three formal sources (i.e *books and/or research papers only) that discuss holistic approach which looks beyond the snubber component and addresses the root cause of the risk. Summrize the holistic approach mentioned in three sources you identified. [9]

*you can use AccessEngineering platform at https://eleanor.lib.gla.ac.uk/record=b4512480 to search reference material.

Q4. Figure Q4 shows a three-phase inverter working in square-wave mode.

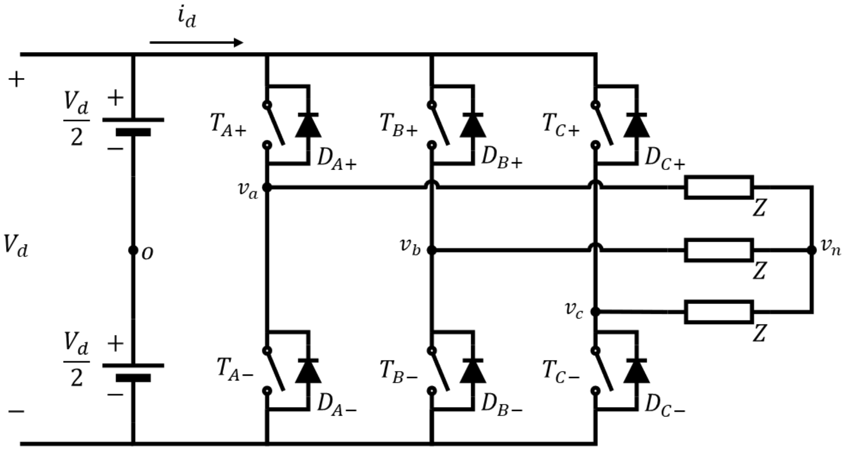

Figure Q4

i. Idenfity the key components of the three-phase inverter shown in Figure Q4 and explain the functions they perform. [3]

ii. A swcithing seuqeson below has been given as follows, compute and tabulate the line voltages. [7]

Seq. #	T_{A+}	T_{B+}	T_{C+}
1	On	Off	On
2	On	On	Off
3	Off	On	Off
4	Off	On	On
5	Off	Off	On
6	On	Off	Off

iii. Assume each step last for 60°, draw the waveform for the line voltages. [6]

iv. Evaluate the waveform and suggest whether this switching sequence is suitable for motor applications. [5]

v. If the inverter output is distorted, explain how this affects the motor's speed control and performance. [4]